INTERNSHIP REPORT_ END OF STUDIES PROJECT
SUBMITTED IN FULFILLMENT OF THE REQUIREMENTS FOR THE
BACHELOR'S DEGREE IN SOFTWARE ENGINEERING AND INFORMATION SYSTEMS

INTELLIGENT E-COMMERCE PRODUCT MANAGER & CHATBOT

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Carried out at :

CAMPUS NUMÉRIQUE DES MÉTIERS

PLAN

01 General introduction

02 Problem statement

03 Proposed solution

04 System Design

05 Implementation

06 Conclusion & perspectives

Chapter 1

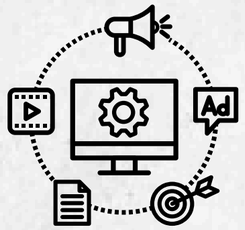
INTRODUCTION

The host organization :

Main Areas of Expertise

- IT Consulting
- Web and Mobile Development
- UI/UX and Graphic Design

- Digital Advertising
- Technical Support and Assistance
- Digital Marketing Solutions

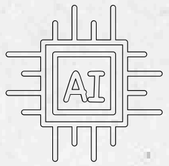


Campus Numérique des Métiers

• CONTEXT

The observation :

Growth of e-commerce worldwide.



AI is becoming more important in digital platforms.

Businesses increasingly sell products online.

Customers expect fast and intelligent services.

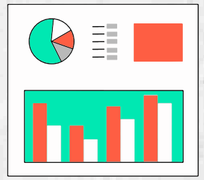
Chapter 2

PROBLEM STATEMENT

• PROBLEM STATEMENT

Manual product management.

Limited customer assistance.



Lack of intelligent business insights.

Research question :

How to design a smart e-commerce platform that streamlines business operations, improves customer satisfaction, and provides intelligent assistance through AI-powered features?

• EXISTING SOLUTIONS STUDY

Features	Shopify	WooCommerce
Product Management	✓	✓
Order Management	✓	✓
Customer Management	✓	✓
Analytics Dashboard	✓	✓
AI-Powered Content Generation	✗	✗
AI Chat Assistant	✗	✗
Return Request Management	Limited	Plugin Required
Real-Time Customer Support	Limited	Plugin Required

Chapter 3

PROPOSED SOLUTION

• PROPOSED SOLUTION

To address the limitations of traditional e-commerce platforms, we developed ShopMind, a smart e-commerce platform that combines conventional online shopping functionalities with AI-powered features

SHOP MIND

Smart Shopping Made Easy

• OBJECTIVES

SHOP MIND
Smart Shopping Made Easy

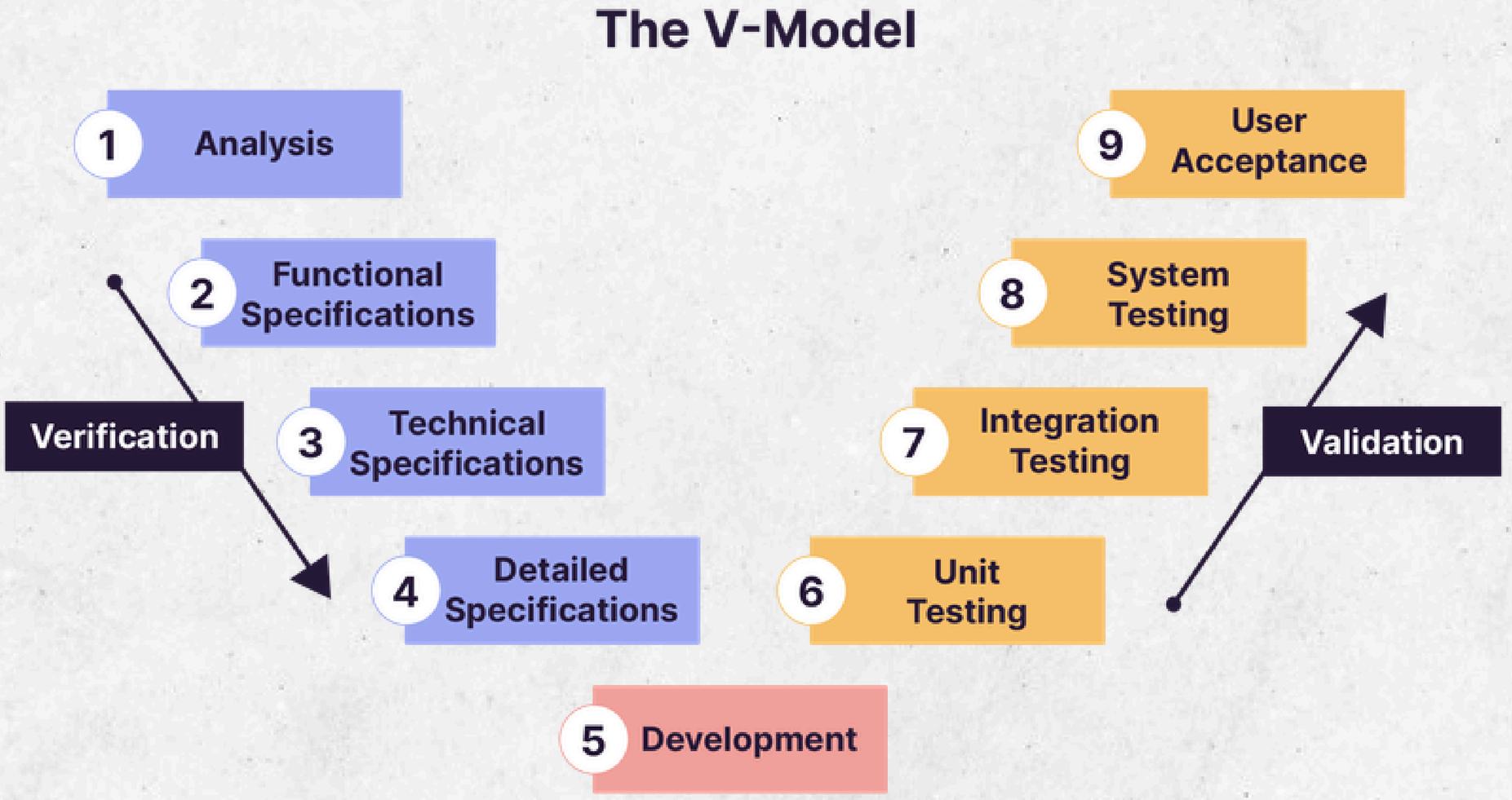
Main objectives :

1. Simplify and accelerate the management of products, orders, and customers.
2. Reduce manual work through intelligent automation.
3. Support decision-making through analytics and business insights.
4. Improve customer experience through intelligent assistance, return management, and communication tools.
5. Deliver a secure, scalable and user-friendly e-commerce platform.

Chapter 4

SYSTEM DESIGN

• METHODOLOGY



• ARCHITECTURE & TECHNOLOGIES

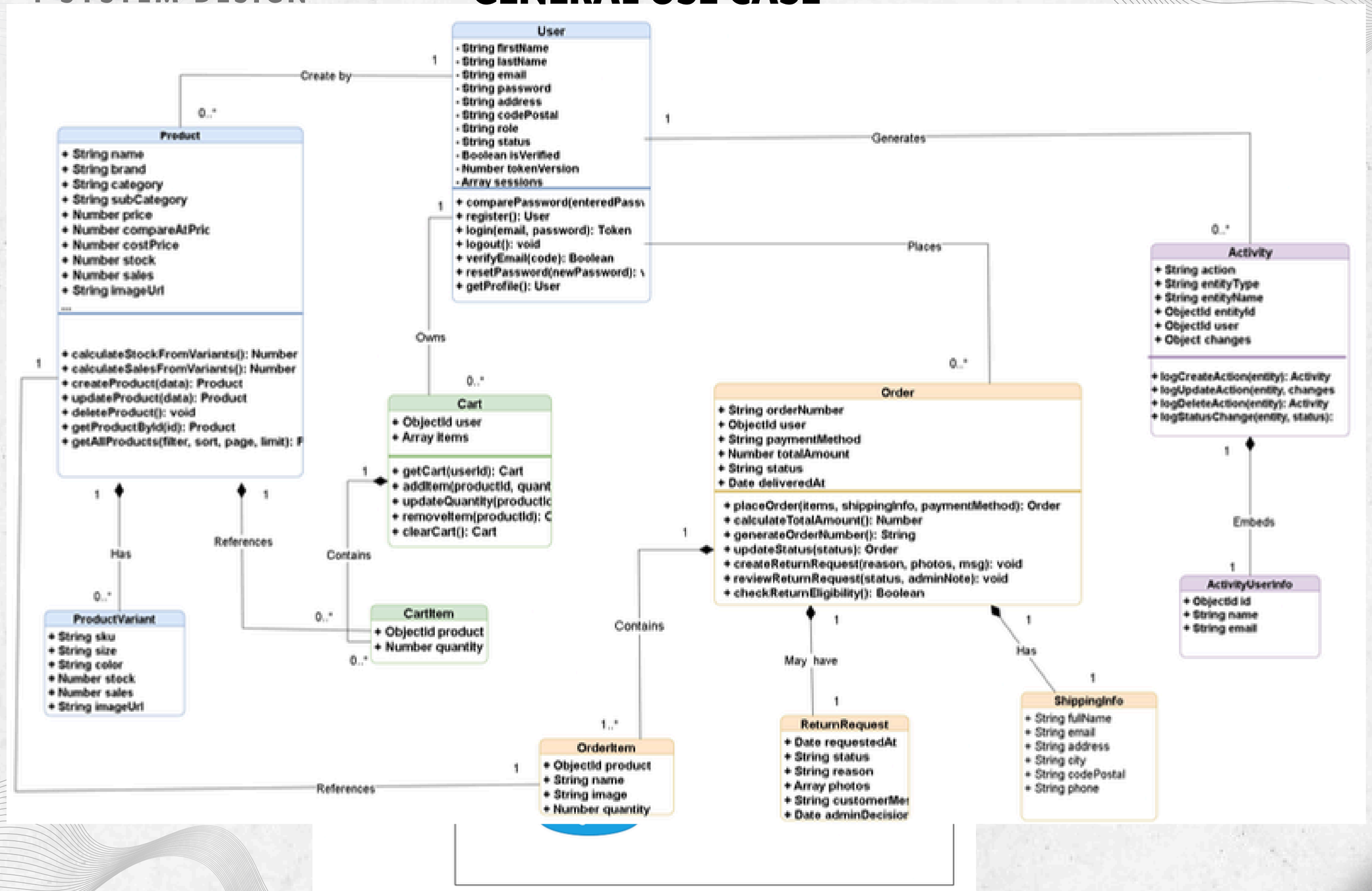
THREE-TIER ARCHITECTURE

Tier	Description
Presentation Tier	User interface and interactions.
Business Logic Tier	Application logic and API processing.
Data Tier	Data storage and management.

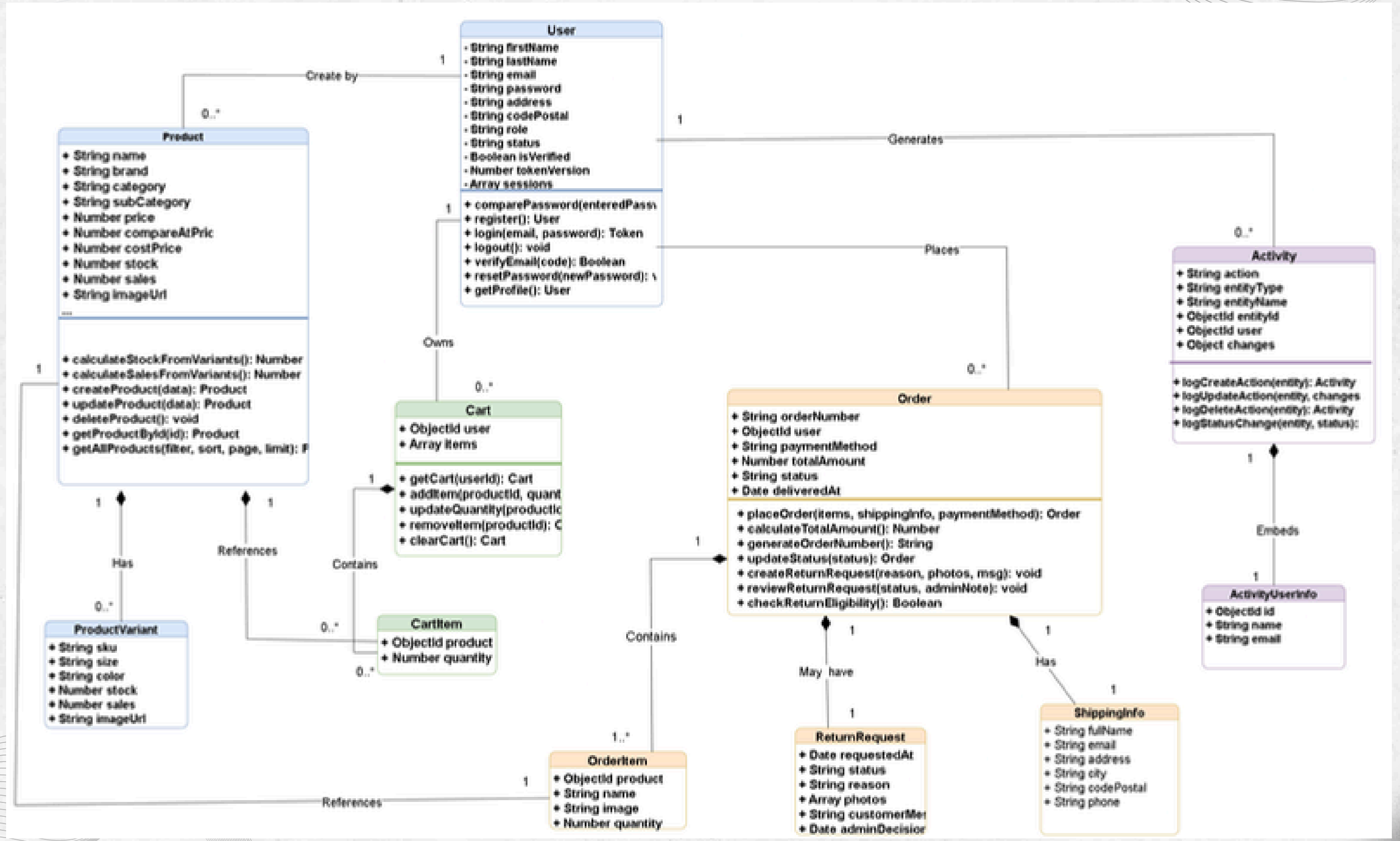
TECHNOLOGIES

Layer / Category	Technologies
Frontend	Angular 17
Backend	Node.js, Express.js
Database	MongoDB Atlas, Mongoose
Authentication & Security	JWT, bcrypt
Communication & Integration	Socket.IO, Nodemailer, CORS
Artificial Intelligence	Puter.js

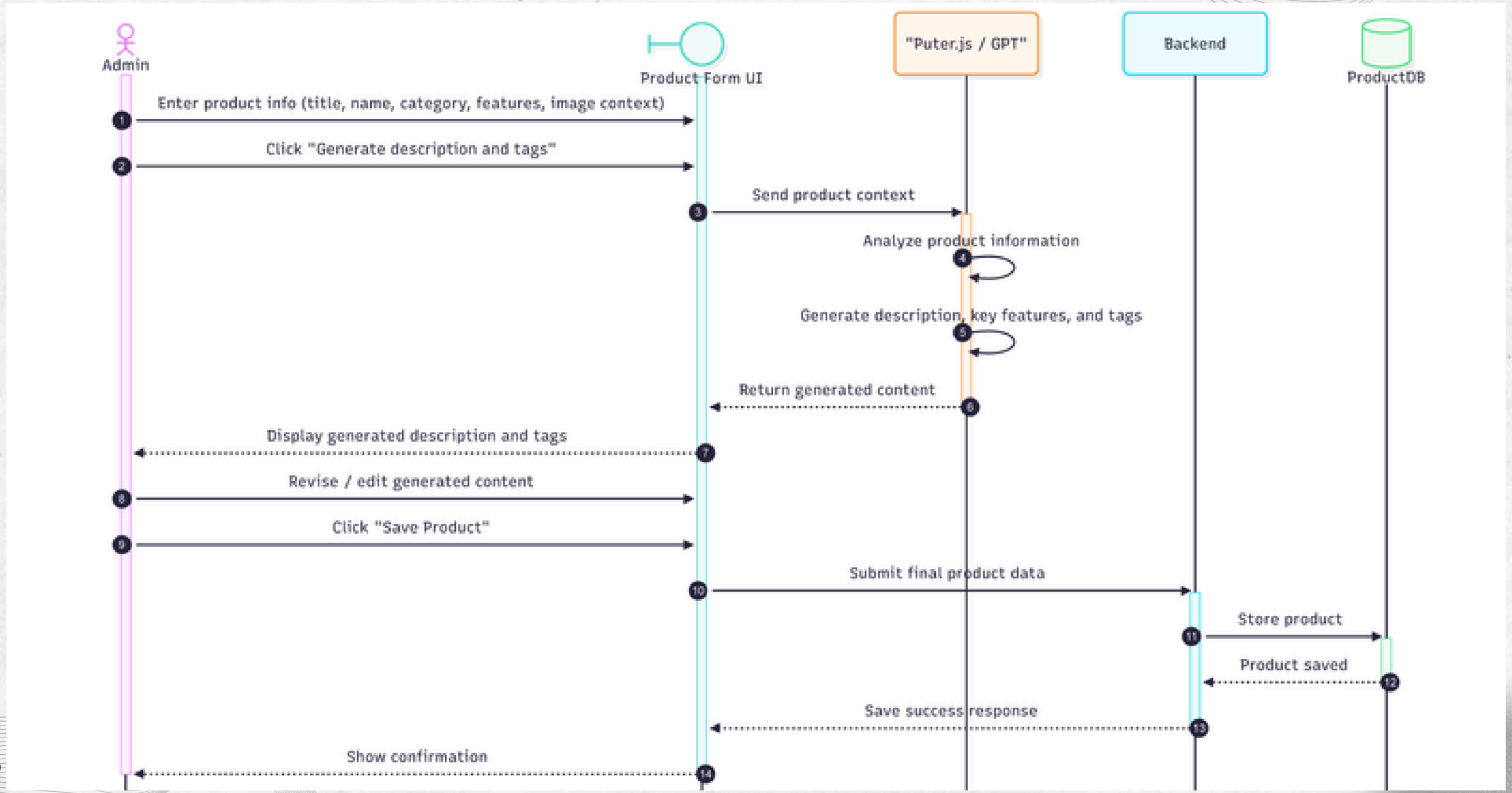
• GENERAL USE CASE



• GENERAL CLASS DIAGRAM

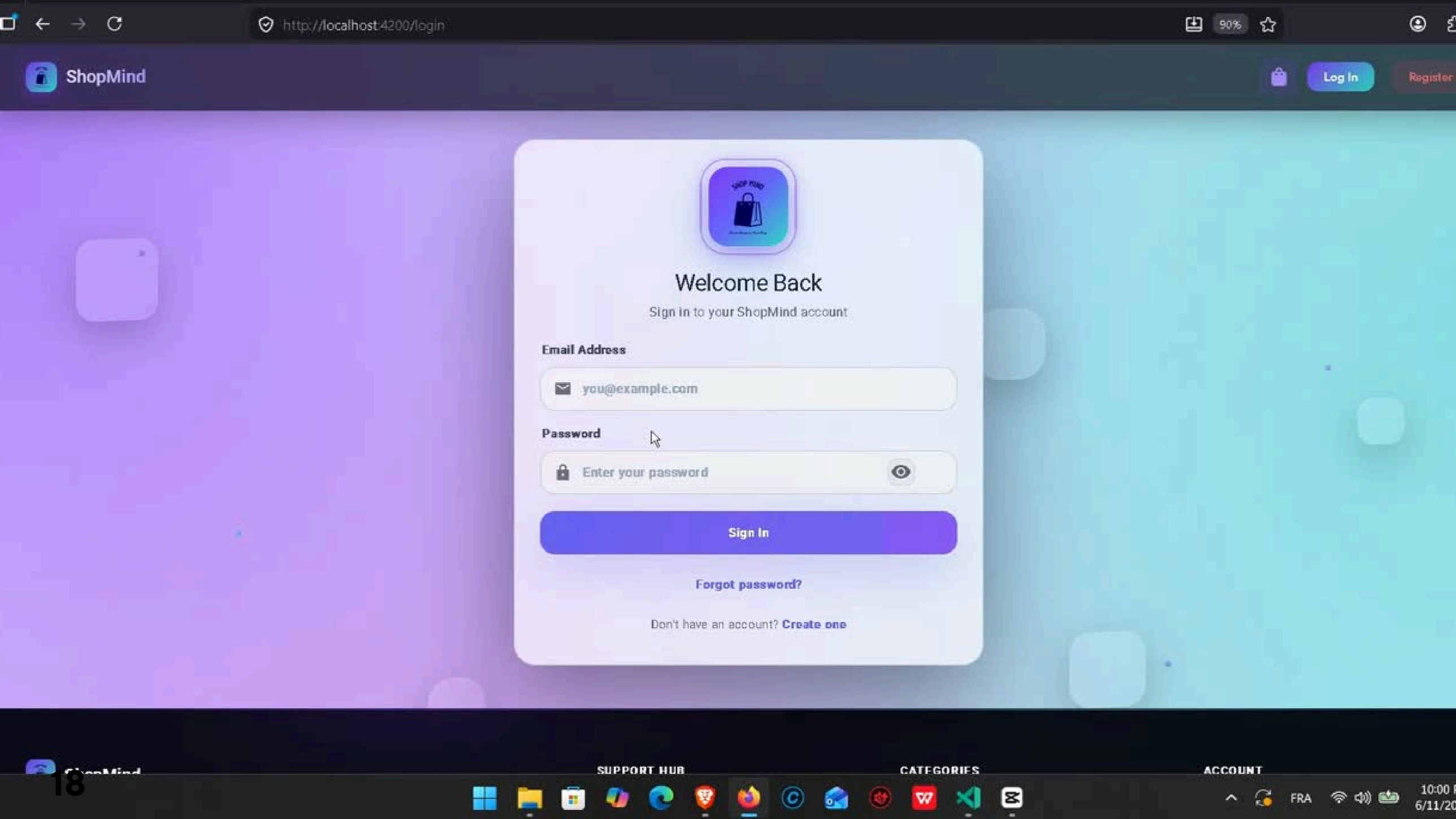


• AI PRODUCT GENERATION SEQUENCE DIAGRAM



Chapter 5

IMPLEMENTATION


http://localhost:4200/login
90%
ShopMind
Log In
Register
Welcome Back
Sign in to your ShopMind account
Email Address
you@example.com
Password
Enter your password
Sign In
Forgot password?
Don't have an account? Create one
SUPPORT HUB
CATEGORIES
ACCOUNT
FRA

Chapter 6

CONCLUSION & PERSPECTIVES

Conclusion :

- Developed ShopMind, a smart AI-powered e-commerce platform.
- Implemented product, customer, order, and return management.
- Integrated AI content generation, chatbot assistance, and analytics features.
- Improved business efficiency and customer experience.

Future Work :

- Integration of online payment gateways.
- Development of a mobile application.
- Implementation of an AI-based recommendation system.
- Advanced analytics and reporting features.
- Support for multiple languages and currencies.

THANK YOU FOR YOUR ATTENTION